Discrete Mathematics – Final Exam

2021 – Moed A

Instructions for students:

1. In the exam there are **10** short questions and **3** open questions. You must answer **all** of them (No choice).
2. Write your solutions in the attached papers only.
3. When asked to write explicitly a "set/function/relation", you must specify all the elements of the set.
4. Use of any external aiding material is **forbidden**. For your convenience, a formula sheet is attached.
5. The duration of the exam is **three hours.**

Good luck!

Part A – Short Questions

- Write your answers clearly in the marked boxes.
- No need to explain how you got to the final answer, **unless it is explicitly asked**.

In counting questions, you can use any of the convenient notations which we used in class, except for the symbol $\sum$ and three dots, '…'.

Question 1 (6 points)

Provide a counterexample to the following claim:

For every 3 sets A,B,C: If $A \cap B \cap C = \emptyset$, then $(A \setminus B) \cup C = A \cup (C \setminus B)$

$$A = B = \{1\}, \qquad C = \emptyset$$

Question 2 (6 Points)

Simplify the following expression $(p \to q) \to \neg(p \to q)$ such that exactly 2 logical connectives appear in your final result, neither of which is implication. Provide only your final result.

$$p \wedge \neg q$$

Explanation:

$(p \to q) \to \neg(p \to q) \equiv \neg(p \to q) \vee \neg(p \to q) \equiv \neg(p \to q) \equiv p \wedge \neg q$

Question 3 (6 points).

Let $f: P(\mathbb{R}) \to P(\mathbb{N})$ be a function defined by : $f(A) = A \cap \mathbb{N}$.

Determine (no explanation required) if the function is injective, and if it is surjective.

- If it's only injective, present two <u>different</u> left inverse functions (no proof).
- If it's only surjective, present two <u>different</u> right inverse functions (no proof).
- If it's both, present its' unique inverse function (no proof).

The function is surjective but not injective, we present two right inverses:

We define $g_1, g_2: P(\mathbb{N}) \to P(\mathbb{R})$ *as follows:*

$$g_1(S) = S \ , g_2(S) = S \cup \{\sqrt{2}\}$$

Question 4 (6 points)

Present the following expression as a short numeral expression (no Sigma, no variables, no '…'):

$$\sum_{k=0}^{24}\left(k\cdot(k-1)\cdot\binom{24}{24-k}\right)$$

$$24\cdot 23\cdot 2^{22}$$

Explanation: Use a combinatoric proof – Given a class of 24 people, how many ways are there to choose a certain number of children to go on a class trip, and from those people choose a trip manager, and a trip consultant (two different positions).

Question 5 (6 points)

How many subsets of $\{1,2,3,\dots,2n\}$ have at least n + 1 elements?

$$\frac{2^{2n}-\binom{2n}{n}}{2}$$

Explanation: Take all subsets, subtract subsets that have exactly n elements. From symmetry we can deduce that exactly half of the sets we have left satisfy the criterion, so we divide by 2.

Question 6 (6 points)

Let $A=\{1,2,\dots,10\}$. What is the smallest value of n, that guarantees that in any collection of n order relations $R_1,R_2,\dots,R_n$ (not necessarily different) over A there are at least 4 relations with the exact same number of minimal elements?

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Pigeonholes – the number of minimal elements a certain order relation on A has – between 1 and 10 minimal elements so we have 10 holes.

Pigeons – the relations : $R_1,R_2,\dots,R_n$.

By the pigeonhole principle, if n = 31 then we are guaranteed to have $\left\lceil\frac{31}{10}\right\rceil=4$ partial orders with the same number of minimal elements.

Question 7 (6 points)

Bastian, Balthazar, Box, Bilbo and Bilby are standing in a row (5 people), in the order that we named them. How many ways are there to rearrange them in a line such that no one remains in his original position in the row?

$$5! - \binom{5}{1}4! + \binom{5}{2}3! - \binom{5}{3}2! + \binom{5}{4}1! - \binom{5}{5}0!$$

Question 8 (6 Points)

Give an example of a set A, and a relation R on A such that R is not symmetric, not transitive, but does satisfy the following property:

$$\forall x, y, z \in A \ \big((xRy \wedge yRz) \rightarrow (x = z)\big)$$

$$A = \{1,2,3,4\} \ , \ R = \{(1,2), (2,1), (3,4)\}$$

Question 9 – 6 Points.

Given is the following equivalence relation over $\mathbb{N}$:

$$nRm \Leftrightarrow \big((n - m)\text{mod}4 = 0 \vee (n \cdot m) \in \mathbb{N}_{odd}\big)$$

Reminder: $(a)\text{mod } b = 0$ means that dividing a by b leaves a remainder of 0.

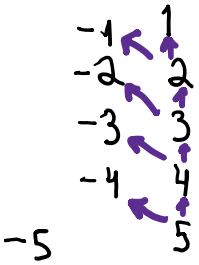
Provide a representation of the quotient set of $\mathbb{N}$ by R.

$$\mathbb{N}/_R = \{[0]_R, [1]_R, [2]_R\}$$

Question 10 (6 points)

Let $A = \{x \in \mathbb{Z} \mid x \neq 0 \wedge |x| \leq 5\}$.
Let R be a strict partial order on A defined as follows: $xRy \Leftrightarrow x > |y|$.
Draw a Hasse diagram for R and specify minimal/maximal elements.

Maximal : 1,-1,-2,-3,-4,-5. Minimal : 5, - 5.



Part B – Open Questions

- Write your answers clearly in the marked boxes. You must prove your answers.
- **Write clear and readable answers! Your grade will also depend on the clarity of your proof and not only the correctness of your answer!**

Question 11 (10 points)

Let p be some real number such that $p \succ 1$.

Prove that for every $n \in \mathbb{N}$ such that $n \geq 1$: $(1+p)^n \geq 1 + np$.

Proof by induction:

Base: *Show the claim holds for n=1 :* $(1+p)^1 \geq 1 + 1 \cdot p$ *– it does hold as both sides are equal.*

Assumption + Step*: Assume the claim holds for some natural* $n \geq 1$*, meaning that* $(1+p)^n \geq 1 + np$*. We will prove the claim for n+1, meaning that* $(1+p)^n \geq 1 + np$*.*

Proof:

$$(1+p)^{n+1} \underbrace{=}_{Exponent\ rules}$$

$$= (1+p)^n \cdot (1+p) \underbrace{\geq}_{Inductive\ assumption}$$

$$\geq (1+np) \cdot (1+p) =$$

$$= 1 + p + np + np^2 >$$

$$> 1 + np + p =$$

$$= 1 + (n+1)p$$

$$= 1 + p + np + np^2 >$$

$$> 1 + np + p =$$

$$= 1 + (n+1)p$$

Question 12 (15 points)

Let R be some relation on A. We define the relation T as follows:

$$aTb \Leftrightarrow \exists_{c \in A}(aRc \wedge cRb)$$

Prove: R is a transitive relation if and only if $T \subseteq R$

$\Rightarrow$

We assume R is transitive and prove that $T \subseteq R$.

- Let $(\mathbf{x}, \mathbf{y}) \in \mathbf{T}$ (meaning xTy).
- By the definition of T there exists $z \in A$ such that $xRz \wedge zRy$.
- By the transitivity of R, we can conclude xRy, meaning $(\mathbf{x}, \mathbf{y}) \in \mathbf{R}$.
- By the definition of containment we have shown $T \subseteq R$.

$\Leftarrow$

We assume $T \subseteq R$ and prove that R is transitive.

- Let $x, y, z \in A$ such that $\mathbf{xRy} \wedge \mathbf{yRz}$.
- By the definition of T we can conclude that xTz, meaning that $(x, z) \in T$.
- Because $T \subseteq R$, we can conclude $(x, z) \in R$, meaning $\mathbf{xRz}$.
- By definition we can conclude R is transitive.

Question 13 (15 points)

Let f, g: A → A be functions. Prove:

a. If f ∘ g ∘ f is invertible, then f is invertible.

b. If f ∘ g ∘ f is invertible, then g is invertible.

a. Assume $f \circ g \circ f$ is invertible. By definition, it is surjective and injective.

From the associativity of composition, we know that:

$f \circ g \circ f = f \circ (g \circ f) = (f \circ g) \circ f$.

So:

- $f \circ (g \circ f)$ is surjective, and by a theorem we have learned it follows that f must also be surjective.
- $(f \circ g) \circ f$ is injective, and by a theorem we have learned it follows that f must also be injective.

f is injective and surjective and is therefore invertible.

b. We have already proven that f is invertible, which means it has an inverse function : $f^{-1}: A \to A$.

Observe the composition of f^{-1} on $f \circ g \circ f$.

As this is a composition of invertible functions, we can conclude from a theorem we have learned that it is invertible.

We prove that the function which results from this composition is in fact g, which proves g is invertible.

$f^{-1} \circ (f \circ g \circ f) \circ f^{-1} = Associativity$

$= (f^{-1} \circ f) \circ g \circ (f \circ f^{-1}) = Composition\ of\ inverse\ functions$

$= i_A \circ g \circ i_A = Identity\ is\ neutral\ in\ composition$

$= g$

Discrete Mathematics – Formula sheet for the exam

Set Theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$
$$A \times B = \{(a,b)|\ a \in A\ and\ b \in B\}$$
$$P(A) = \{S|S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$
$$|P(A)| = 2^{|A|}$$

De-Morgan:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad A\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$
$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad A\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions/Relations:

$$i_A = \{(a,a)|a \in A\}$$

Logic:

For all formulas α, β, γ :

De-Morgan:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$
$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

Distributive:

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$
$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

Negation rules:

$$\neg\forall x(\alpha) \equiv \exists x(\neg\alpha)$$
$$\neg\exists x(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k items out of n different items:

	With repetition	Without repetition
With order	n^k	$\frac{n!}{(n-k)!}$
Without order	$\binom{k+n-1}{n-1}$	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$

Binomial Formula

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial Coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \cdots + i_k = n$$

Multinomial Formula:

$$(x_1 + x_2 + x_3 + \cdots + x_k)^n = \sum_{i_1+i_2+i_3+\cdots+i_k=n} \binom{n}{i_1, i_2, i_3, \dots, i_k} \cdot x_1^{i_1} \cdot x_2^{i_2} \cdot x_3^{i_3} \cdot \dots \cdot x_k^{i_k}$$

Pascal's Triangle:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

$$\binom{n}{k} = \binom{n}{n-k}$$

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Inclusion-exclusion principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$

Generalized Pigeonhole Principle:

If we put m pigeons in n pigeonholes, then there is at least one pigeonhole in which there will be at least $\left\lceil \frac{m}{n} \right\rceil$ pigeons.